Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/20/2019
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Project Title:	Comprehensive Whole Genome Analysis of Differential Epigenetic Effects of Space Travel on Monozygotic Twins		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/14/2014	**End Date:**	12/31/2018
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 12/31/2018 per NSSC information (Ed., 6/20/18) NOTE: End date changed to 4/30/2018 per NSSC information (Ed., 3/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Several substantial medical barriers to long-term space travel include radiation damage, weightlessness, and stress caused by the specialized spacecraft environment. This Human Exploration Research Opportunities (HERO) grant solicitation requests applicants to take advantage of the unusual opportunity to compare identical twins, one of whom will be traveling in orbit for a year while the other will remain on Earth. The area of science that most directly addresses this opportunity is epigenetics, the study of chemical changes to DNA and the associated proteins, other than the DNA sequence itself. Our laboratory is the leading center that has pioneered the methods for analyzing epigenetics at a genome-scale level, termed epigenomics, and for applying these methods to the discovery of epigenetic causes and mediators of common human disease. These discoveries include the first analysis of epigenomic changes in individuals over time, the first complete epigenomic map of human cancer, the discovery of an epigenomic basis for autoimmune disease, and epigenomic mechanisms for injury response. In this proposal, we will test the hypothesis that space-induced exposure causes epigenome damage. We will determine the nature of this damage, and whether it is correctible spontaneously. We will measure three epigenome-related modalities in identical twins, namely: DNA methylation, a chemical change in the base cytosine; large heterochromatin regions related to chemical modifications of histones; and their functional relationship to gene expression of RNA (expression data generated by other Twins' investigators). We will obtain measurements prior to launch, after reentry, and at quarterly intervals in space, as well as after any unforeseen environmental exposure such as a spike in radiation. We will complement this analysis with whole genome DNA sequencing data generated by other Twins' researchers before and after the flight on both twins, in order to relate epigenomic damage to subsequent DNA damage. Given the potentially reversible nature of epigenetic changes, this work may help to identify or mitigate exposure to radiation, weightlessness, and environmental stressors, which are considered the greatest barriers to long term space flight. Thus our work could have a huge impact on long-duration space travel, if we can identify potentially reversible epigenetic precursors of genetic damage to astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The Principal Investigator (PI) and his staff have spent considerable effort optimizing protocols for cell fractionation, DNA and chromatin analysis and sequencing. From blood samples collected from each twin throughout the study, we have isolated several classes of immune cells (CD4, CD8, and CD19 positive lymphocyte cells) for epigenome analyses. Helper T cells (CD4+) work together with killer T cells (CD8+) to combat and destroy infected or cancerous cells. B cells (CD19+) make antibodies to respond to these infections. We have also isolated DNA from buccal and saliva samples that await sequencing. We have performed whole genome bisulfite sequencing (WGBS) on CD4+ and CD8+ cells from 4 time points in each twin, to measure DNA methylation at a genome-wide scale. Novel mathematical modeling tools we developed allowed us to examine paired samples to identify regions of the genome where the epigenome differs in a meaningful way. Using these methods, we have interrogated our data for epigenetic changes that arose during Scott Kelly's time in space and returned (or not) to baseline levels upon his return. Globally, these changes were minimal when compared to those in Mark Kelly over the same time period. Therefore, we conclude that gross epigenetic changes detected in CD4+ and CD8+ cells from Scott are well within the "normal" range of variation seen in his earth-bound twin. That being said, we have characterized these small methylation changes in Scott and these are summarized below: • CD4+ and CD8+ samples analyzed via WGBS: o Mark – pre-flight (12/3/14), in-flight (7/30/15 and 1/20/16), post-flight (7/17/16); o Scott – pre-flight (10/16/14), in-flight (6/11/15 and 12/12/15), post-flight (6/14/16); • We found subtle, but global differences in genome-wide DNA methylation: o Methylation is very similar in twins pre-flight. o In CD4+ cells, Scott's global DNA methylation is decreased in-flight and returns to near baseline upon return. o In contrast, there is a slight increase in global methylation in CD8+ samples collected in-flight that returns to baseline. o Mark's global DNA methylation level showed the greatest increase from the first to the second "in-flight" time point in both CD8+ and CD4+ samples (2.4% and 2.7% increase, respectively) before returning to near baseline. o Visualization of all gene promoters revealed no detectable changes in methylation throughout the duration of the study for either subject. o To identify particular genes displaying differential epigenetic regulation, we calculated the Jensen-Shannon distance (JSD) among all possible pairwise comparisons to quantify the epigenetic discordance between samples. Large Jensen-Shannon distances indicate that, although there may only be small changes in mean methylation levels, there are pronounced changes in the probability distributions of methylation levels in these regions. For Scott, two example genes, COL1A1 and SLC1A5, have high JSD in their promoters as well as at locations throughout their gene bodies, particularly when comparing pre- or post-flight time points to the second in-flight time point. o Given the lack of biological replicates, we cannot ascribe statistical significance to individual loci, but we have performed gene ontology analysis of genes ranked by their promoter JSD and found statistically significant enrichments in categories related to platelet aggregation, and response to UV-B in CD4+ cells, somatostatin signaling in CD8+ cells, and T-cell differentiation in CD4+ and CD8+ cells. These categories were largely distinct from those enriched in the ground-based twin, suggesting that while DNA methylation patterns were altered in both subjects, these changes occurred at distinct loci in each subject. Discussion: We cannot ascribe clinical significance to these changes in DNA methylation or relate to a known human health effect as DNA methylation is not a CLIA-approved diagnostic assay. However, these changes could help in understanding the effects of future space flight by indicating which genes or regions of the genome are most responsive to the space environment. Our data analysis methods are somewhat unique in that we are also able to assess biological noise, or "entropy" in the epigenome, a measure of lability in gene activity. We found changes in entropy between the twins, as well. While again not of known clinical significance, these results could indicate genes that are more sensitive to a changing environment, whether on Earth or in space. We have coordinated with data managers with Mike Snyder's group to handle all data uploads so that all Twins' data can be easily accessed by all the Twins' researchers and, upon acceptance of our Twins' Study manuscript, these data will be made available through NASA's Life Sciences Data Archive (LSDA) upon approval by NASA.

Bibliography Type:	Description: (Last Updated: 05/20/2019)
Articles in Peer-reviewed Journals	Garrett-Bakelman FE, Darshi M, Green SJ, Gur RC, Lin L, Macias BR, McKenna MJ, Meydan C, Mishra T, Nasrini J, Piening BD, Rizzardi LF, Sharma K, Siamwala JH, Taylor L, Vitaterna MH, Afkarian M, Afshinnekoo E, Ahadi S, Ambati A, Arya M, Bezdan D, Callahan CM, Chen S, Choi AMK, Chlipala GE, Contrepois K, Covington M, Crucian BE, De Vivo I, Dinges DF, Ebert DJ, Feinberg JI, Gandara JA, George KA, Goutsias J, Grills GS, Hargens AR, Heer M, Hillary RP, Hoofnagle AN, Hook VYH, Jenkinson G, Jiang P, Keshavarzian A, Laurie SS, Lee-McMullen B, Lumpkins SB, MacKay M, Maienschein-Cline MG, Melnick AM, Moore TM, Nakahira K, Patel HH, Pietrzyk R, Rao V, Saito R, Salins DN, Schilling JM, Sears DD, Sheridan CK, Stenger MB, Tryggvadottir R, Urban AE, Vaisar T, Van Espen B, Zhang J, Ziegler MG, Zwart SR, Charles JB, Kundrot CE, Scott GBI, Bailey SM, Basner M, Feinberg AP, Lee SMC, Mason CE, Mignot E, Rana BK, Smith SM, Snyder MP, Turek FW. "The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight." Science. 2019 Apr 12;364(6436):eaau8650. https://science.sciencemag.org/content/364/6436/eaau8650.long ; PubMed PMID: 30975860 , Apr-2019